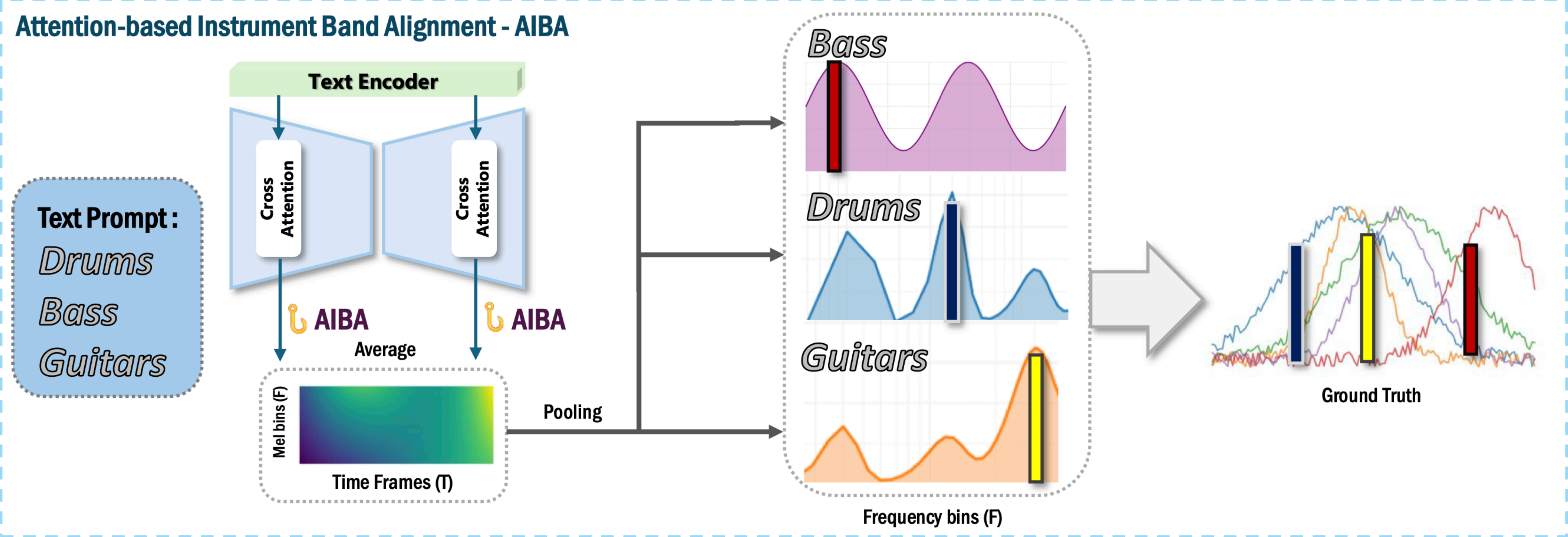
AIBA: ATTENTION-BASED INSTRUMENT BAND ALIGNMENT FOR TEXT-TO-AUDIO DIFFUSION

Junyoung Koh†, Soo Yong Kim, Yongwon Choi, Gyu Hyeong Choi

solbon1212@yonsei.ac.kr

Architecture Visualization



AIBA: Attention-to-Mel Mapping for TTA

We present **AIBA (Attention-In-Band Alignment), a lightweight, training-free pipeline to quantify where text-to-audio diffusion models attend on the time–frequency (T–F) plane**. AIBA (i) hooks into cross-attention during inference to record attention probabilities; (ii) projects them to fixed-size mel grids that are directly comparable to audio energy; and (iii) scores agreement with instrument-band ground truth via interpretable metrics (T–F IoU/AP, frequency-profile correlation, pointing game).

On Slakh2100 [1] with an AudioLDM2 [2] backbone, AIBA reveals consistent instrument dependent trends (e.g., bass favoring low bands) and achieves high precision with moderate recall. Our code enables reproducible extraction, mapping, and evaluation at scale.

Visualizing Attention Maps through Cross-Attention Hooking

We attach processors to every cross-attention block (attn2) in the AudioLDM2 U-Net. At each call, we compute

$$A = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right), \quad A \in \mathbb{R}^{H\times T_q\times T_k},$$

$$\bar{A} = \frac{1}{H}\sum_{h=1}^{H} A_h \;\in\; \mathbb{R}^{T_q\times T_k},$$

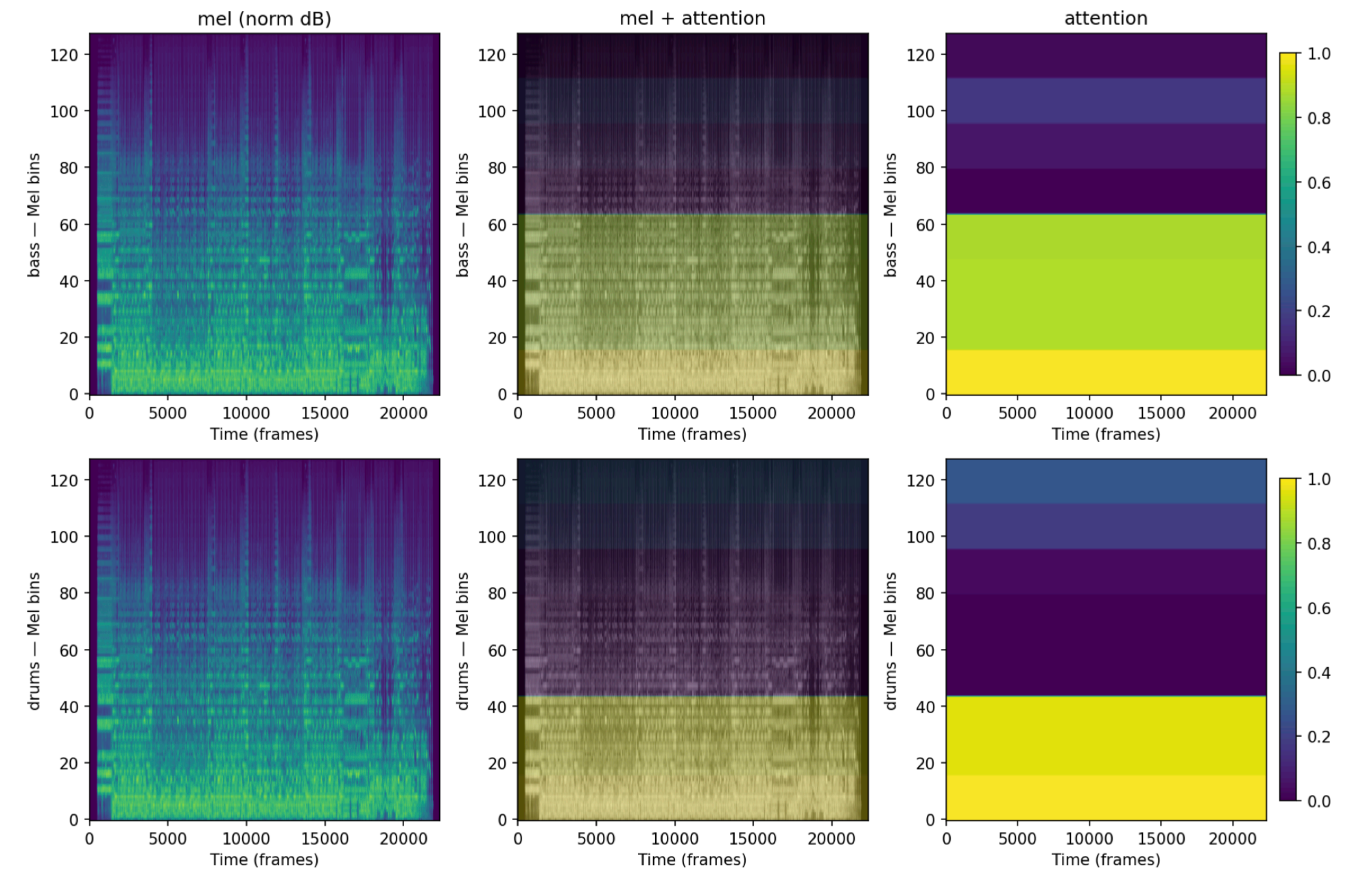
where $Q \in \mathbb{R}^{H\times T_q\times d_k}$ and $K \in \mathbb{R}^{H\times T_k\times d_k}$ are per-head query/key projections, H is the number of heads, and T_q, T_k denote query/key token lengths. We then average over heads,

$$p_{\mathrm{mean}}(t) = \frac{1}{T_k}\sum_k \bar{A}(t,k),$$

Results on Slakh2100 Dataset

Metric	Micro	Macro	Pos.-weighted
Precision	0.984	0.984	0.984
Recall	0.755	0.756	0.755
F1	0.854	0.848	0.848
IoU	0.746	0.746	0.746

AIBA achieves high micro-precision (0.98) and F1 (0.85), indicating that attention consistently overlaps with ground-truth bands. Recall is lower (0.75), suggesting that while attended regions are correct, some ground-truth energy is missed. Macro- and class-weighted scores are similar, showing balanced performance across instruments.



Time-Frequency mel-spectrogram

Reference

[1] Ethan Manilow, Gordon Wichern, Prem Seetharaman, and Jonathan Le Roux. **Cutting music source separation some Slakh: A dataset to study the impact of training data quality and quantity**. In Proc. IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA). IEEE, 2019.

[2] Haohe Liu, Zehua Chen, Yi Yuan, Xinhao Mei, Xubo Liu, Danilo Mandic, Wenwu Wang, and Mark D Plumbley. **AudioLDM: Text-to-audio generation with latent diffusion models.** Proceedings of the International Conference on Machine Learning, pages 21450–21474, 2023.

We are looking for …

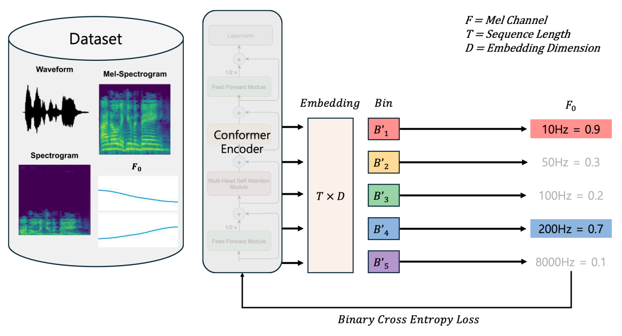
The goal of our **M-A-A-P Lab** is to advance Music AI research and collaboration across academia and industry. We are looking for motivated researchers and supporters to join and contribute to our journey.

In particular, I'm looking for AI researchers interested in **Physical AI and Music**, such as dancing robots or other music-driven interactive systems.

More Research to Share!

Our Paper : AIBA

Jamendo QA Dataset



Multi Pitch Estimation

Myna-RPE SSL Learning

Multi-Hop Music QA Dataset

Audio Generation with RLHF